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Subject: - Telecommunication Project

Report Topic: - Assignment I

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INTRODUCTION

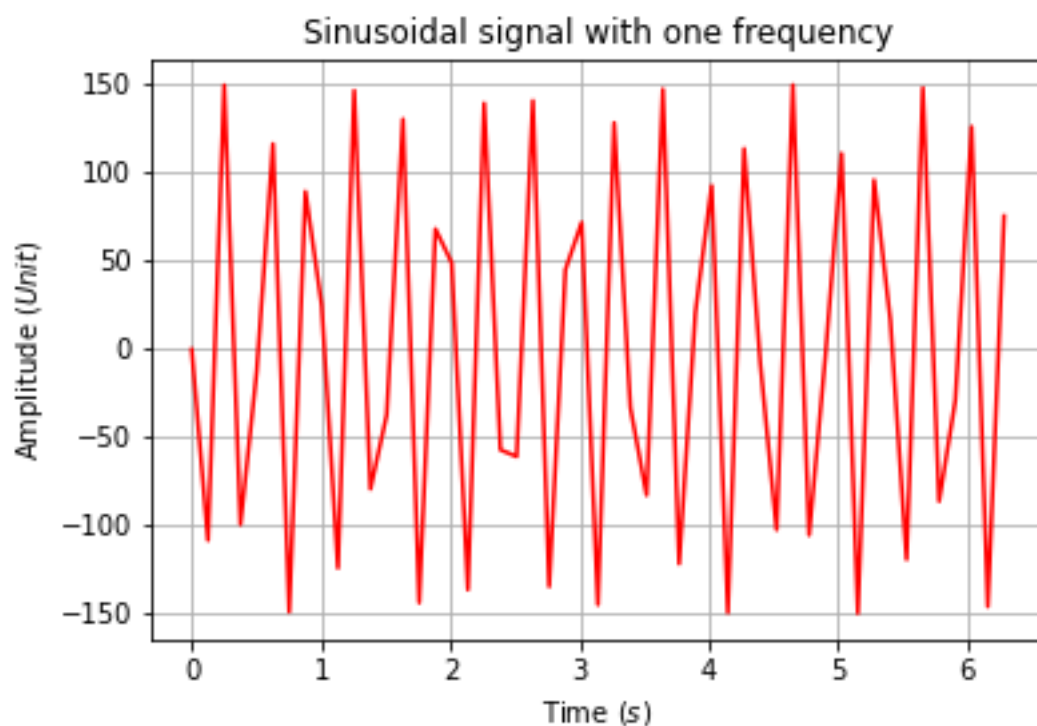
Time series of a periodic signal:

A periodic signal is one that precisely repeats the sequence of values after a fixed length of time, known as the period.

Example: Sinusoidal signal, cosine signal.[\(1\)](#)

A sinusoidal signal is a periodic signal that oscillates (moves up and down) on a periodic basis is known as a sinusoidal wave signal. In one complete cycle, the geometric waveform of a sinusoidal signal generates an S-shaped wave.

Frequency is the number of times a sinusoidal signal completes a cycle in one second. It displays how quickly the signal is changing over time. Amplitude which represents the sinusoidal signal's highest value. It is also called peak value or maximum value.[\(2\)](#)



Here, we are generating the long time series of a sinusoidal signal with a frequency 5Hz and an amplitude of 150, sampled at a rate of 51.

Window function or apodization function or tapering function:

To measure the frequency component of a signal, FFT is used which assumes a continuous spectrum. When windows are created, discontinuities have been created that result in high-frequency components compared to the original signal. The primary objective of window functions is tapering.[\(3\)](#)

Factors describing window function are:

- The width of the main lobe, or the frequency bin at which the power is equal to half the maximum response.
- Attenuation of the side lobes.

Window functions have different distortion characteristics which depend on the selected window function.

Types of window functions are:

- Rectangular window
- Hann window
- Hamming window
- Blackman window
- Gaussian window
- Bartlett window
- Kaiser window
- Flat top window

FFT:

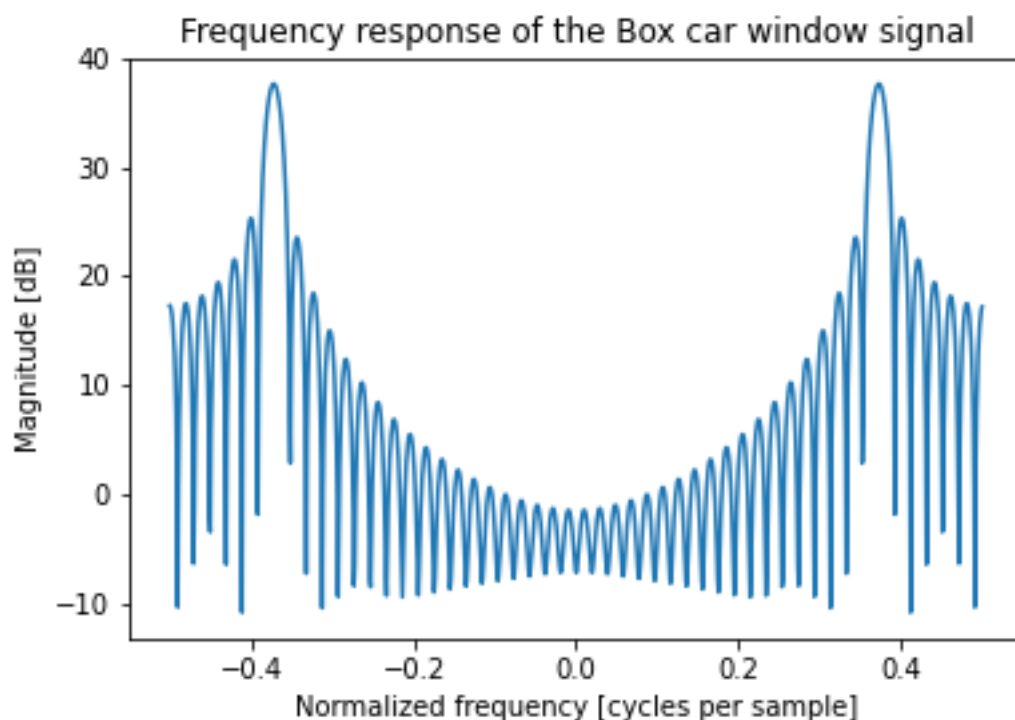
A fast Fourier transform (FFT) is an algorithm that computes the discrete Fourier transform (DFT) of a sequence, or its inverse (IDFT). A signal is transformed by Fourier analysis from its original domain, which is frequently time or space, to a representation in the frequency domain, and vice versa. [\(4\)](#)

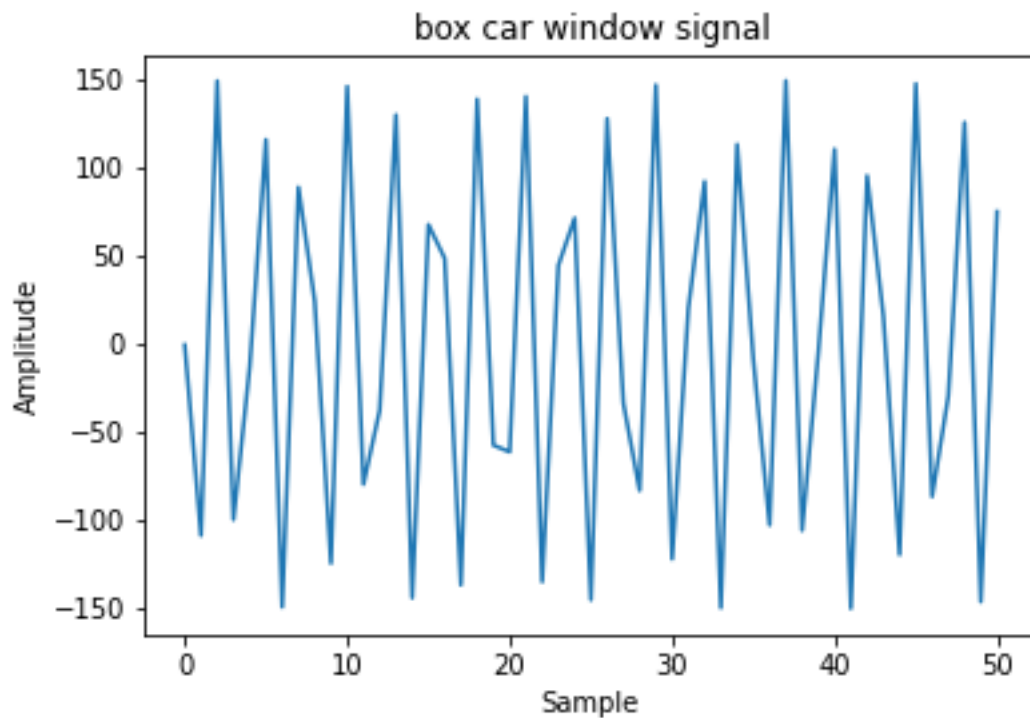
For a longer FFT length, greater frequency resolution and enough computational resources are required. When a high-frequency resolution is not required and computing resources are to be conserved, a shorter FFT length is observed.

Here we are discussing only a few window functions, which are used in the task.

Rectangular window or Boxcar window:

For better frequency resolution, the length of window function is increased which results in narrowing of the main lobe. The height of the side lobes is unaffected by the window's length. The windowing capabilities of rectangular windows are subpar.



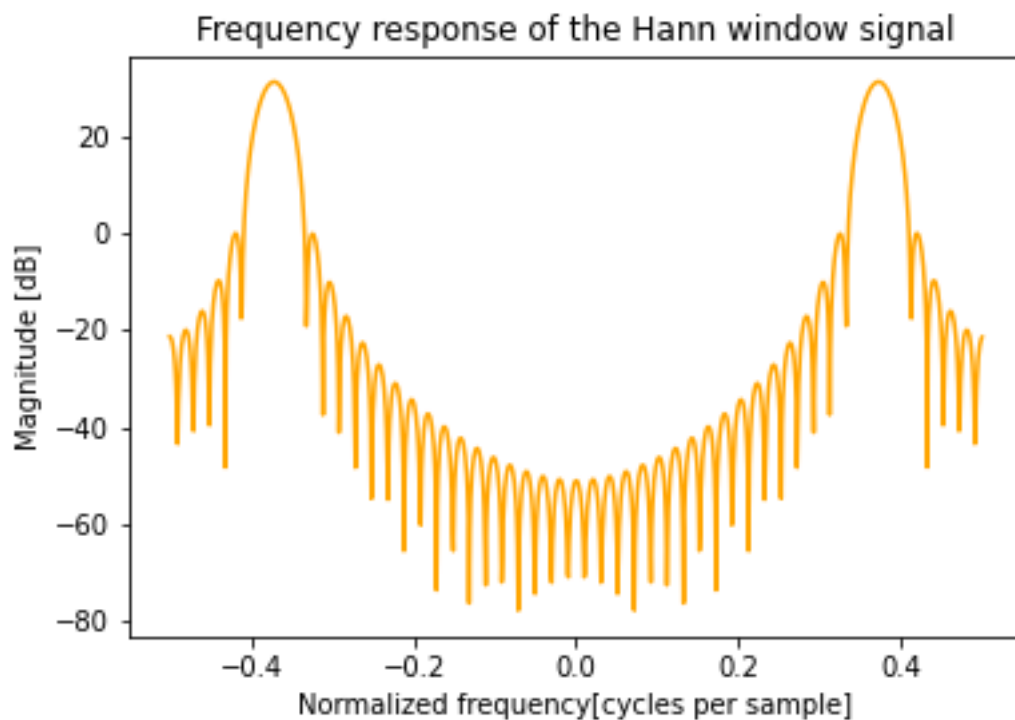
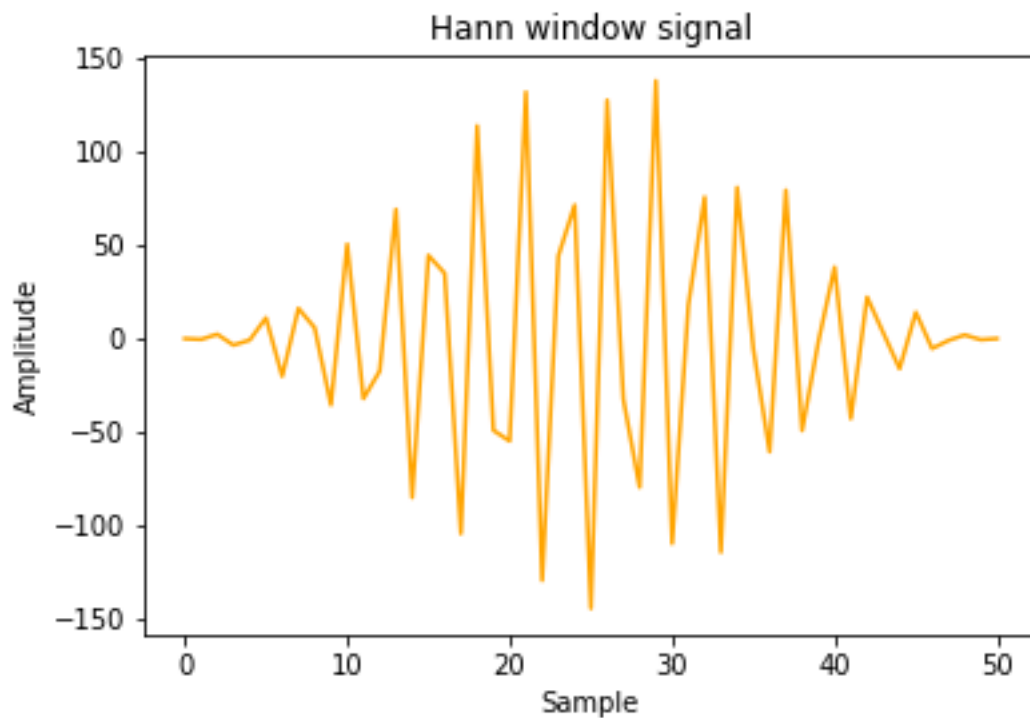


In time domain the windowed signal looks like sinusoidal signal, where as in the spectral domain, we can see two large spikes in the middle of signal. Where as the complete signal has very less amplitude.

Hann window or cosine bell:

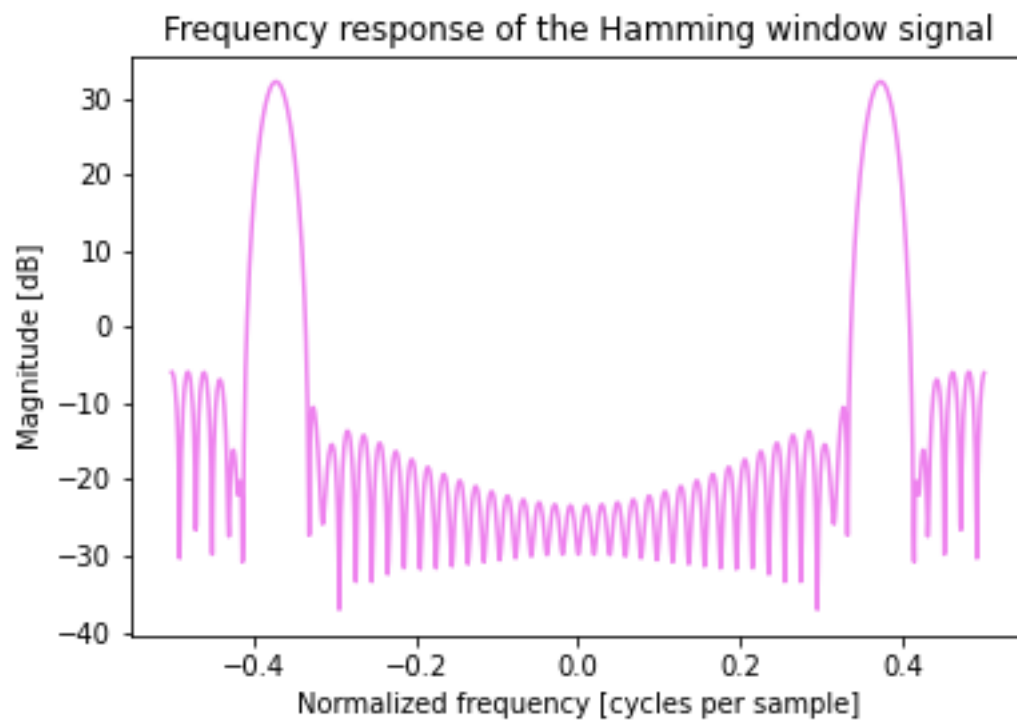
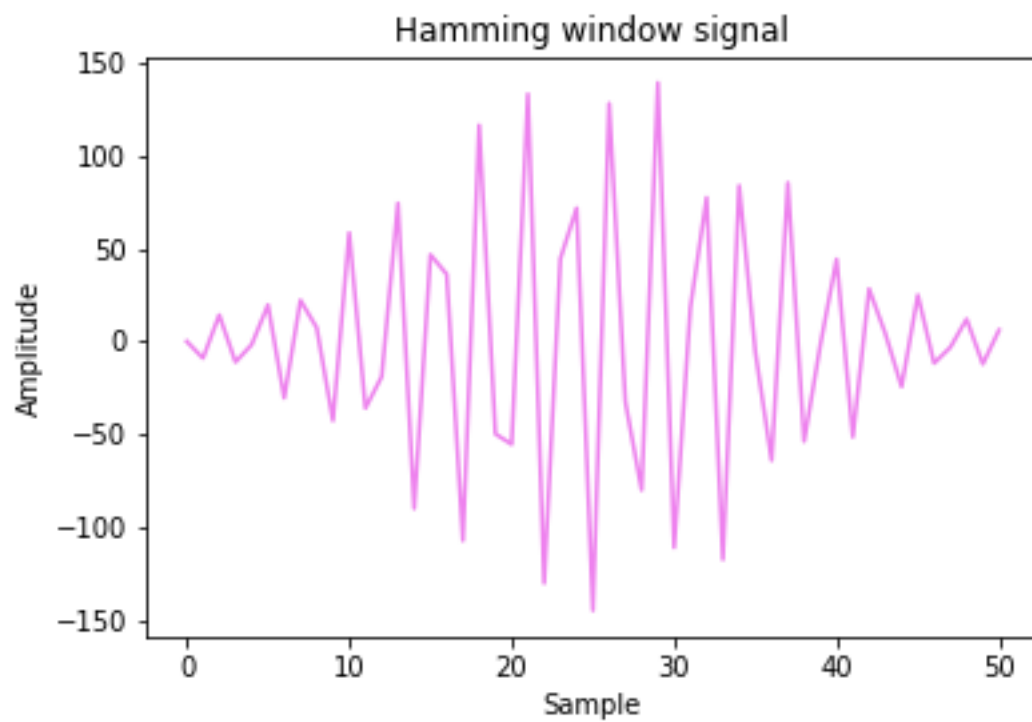
The Hann window is the taper produced by a weighted cosine.

The Hann window touches zero on both ends, erasing any discontinuity. Hann window looks like elevated Gaussian (just a little bit broader). It has good frequency resolution and reduced spectral leakage.



Hamming window:

The Hamming window is the taper produced by a weighted cosine. There is a tiny signal discontinuity because the hamming window does not quite reach zero. Narrow the distribution, lesser the energy captured. Hamming window maintains good spectral energy.

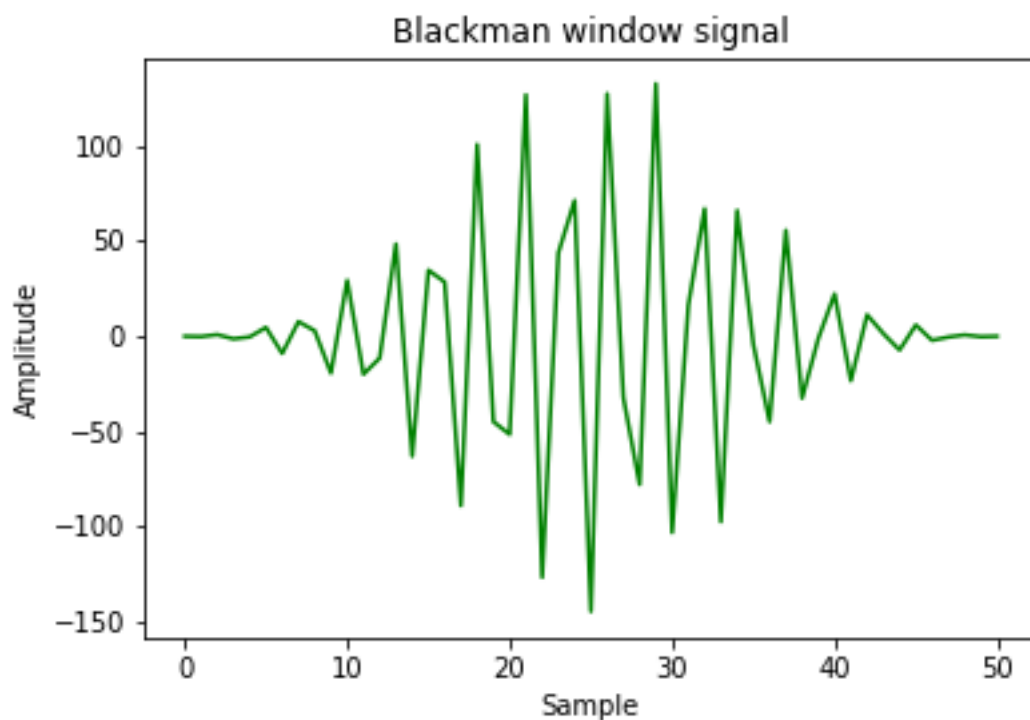


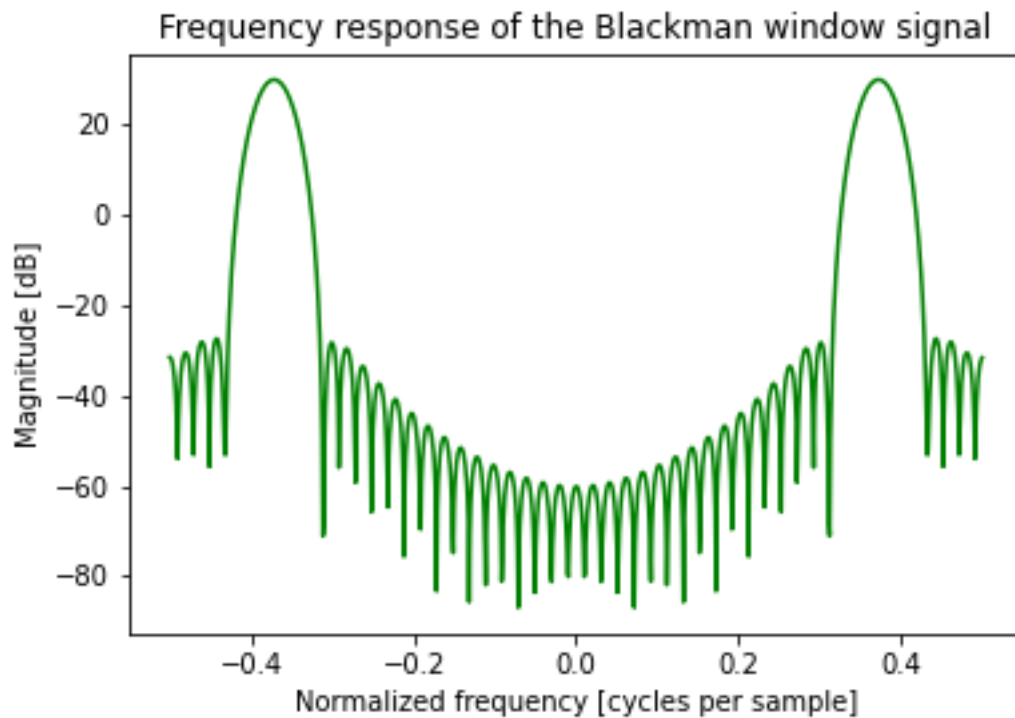
Blackman window

The first three terms of a summation of cosines are used to generate the taper known as the Blackman window. The spectrum that results from this has a broad peak but good side lobe compression. It is symmetric.

In the Hann, Hamming, and Blackman windows, values are smoothed. The process of removing the foot is called apodization. The sampled signal's discontinuities at the beginning and finish are smoothing.

The flat top window gives the exact amplitude of the signal and a rectangular window gives the exact frequency of the peaks of a signal. [\(5\)](#)





Comparing the advantages and disadvantages of Blackman, hamming, and hanning windows:

In comparison to Hanning, Hamming has higher side lobes but a worst side lobe roll-off rate. A hybrid of the other two windows, the Blackman has a higher side lobe height than either window and a better side lobe roll-off rate than the Hanning. The primary lobe width of the Blackman becomes wider.

Appendix: [6](#)

```
import numpy as np
import matplotlib.pyplot as plt
from numpy.fft import fft, fftshift
from scipy.signal import get_window

#-----
#####Sinusoidal signal#####
npts=51
t = np.linspace(0, 2*np.pi, npts, endpoint=True) # time vector

f0 = 5.0 # Frequency in Hz
#f1=18
A1 = 150.0 # Amplitude in Unit
#A2 = 70.0 # Amplitude in Unit
y1 = A1 * np.sin(2*np.pi*f0*t) # sinusoidal signal
#y2 = A2 * np.sin(2*np.pi*f1*t) # sinusoidal signal
#y3= y1+y2

plt.figure(1)
plt.plot(t,y1,"red")
#plt.plot(t,y2)
```

```
plt.title("Sinusoidal signal with one frequency")
plt.xlabel('Time ($s$)')
plt.ylabel('Amplitude ($Unit$)')
plt.grid('on')
plt.savefig('Sinusoidal signal with one frequency.png')
```

```
#-----
##### Hanning window function #####
```

```
window = np.hanning(npts)
windowed_sig = window * y1
```

```
plt.figure(2)
plt.plot(windowed_sig,"orange")
```

```
plt.title("Hann window signal")
plt.ylabel("Amplitude")
plt.xlabel("Sample")
#plt.ylim([0, 1])
plt.savefig('Hann_window signal.png')
```

```
#FFT
A = fft(windowed_sig , 1024) / 50
```

```
mag = np.abs(fftshift(A))
```

```
freq = np.linspace(-0.5, 0.5, len(A))
```

```
with np.errstate(divide='ignore', invalid='ignore'):
```

```
    response = 20 * np.log10(mag)
```

```
    response = np.clip(response, -100, 100)
```

```
plt.figure(3)
```

```
plt.plot(freq, response, "orange")
```

```
plt.title("Frequency response of the Hann window signal")
```

```
plt.ylabel("Magnitude [dB]")
```

```
plt.xlabel("Normalized frequency[cycles per sample]")
```

```
plt.axis('tight')
```

```
plt.savefig('Frequency response of the Hann window signal.png')
```

```
#-----
```

```
##### Hamming #####
```

```
window = np.hamming(npts)
```

```
windowed_sig = window * y1
```

```
plt.figure(4)
```

```
plt.plot(windowed_sig, "violet")
```

```
plt.title("Hamming window signal")
```

```
plt.ylabel("Amplitude")
```

```
plt.xlabel("Sample")
```

```
#plt.ylim([-0.1, 1.1])
plt.savefig('Hamming_window signal.png')
```

```
A = fft(windowed_sig , 1024) / 50
```

```
mag = np.abs(fftshift(A))
```

```
freq = np.linspace(-0.5, 0.5, len(A))
```

```
with np.errstate(divide='ignore', invalid='ignore'):
    response = 20 * np.log10(mag)
```

```
response = np.clip(response, -100, 100)
```

```
plt.figure(5)
plt.plot(freq, response, "violet")
plt.title("Frequency response of the Hamming window signal")
plt.ylabel("Magnitude [dB]")
plt.xlabel("Normalized frequency [cycles per sample]")
plt.axis('tight')
plt.savefig('Frequency response of the Hamming window signal.png')
```

```
#-----
##### Blackman #####
window = np.blackman(npts)
```

```
windowed_sig = window * y1
```

```
plt.figure(6)
```

```
plt.plot(windowed_sig,"green")
```

```
plt.title("Blackman window signal")
```

```
plt.ylabel("Amplitude")
```

```
plt.xlabel("Sample")
```

```
#plt.ylim([-0.1, 1.1])
```

```
plt.savefig('Blackman_window signal.png')
```

```
#FFT
```

```
A = fft(windowed_sig , 1024) / 50
```

```
mag = np.abs(fftshift(A))
```

```
freq = np.linspace(-0.5, 0.5, len(A))
```

```
with np.errstate(divide='ignore', invalid='ignore'):
```

```
    response = 20 * np.log10(mag)
```

```
    response = np.clip(response, -100, 100)
```

```
plt.figure(7)
```

```
plt.plot(freq, response,"green")
```

```
plt.title("Frequency response of the Blackman window signal")
```

```
plt.ylabel("Magnitude [dB]")
plt.xlabel("Normalized frequency [cycles per sample]")
plt.axis('tight')
plt.savefig('Frequency response of the Blackman window signal.png')
```

```
#-----
##### Boxcar #####
```

```
#window = [*np.zeros(10), *np.ones(30), *np.zeros(10)]
window = get_window('boxcar', npts)
windowed_sig = window * y1
```

```
plt.figure(8)
plt.plot(windowed_sig)
```

```
plt.title("box car window signal")
plt.ylabel("Amplitude")
plt.xlabel("Sample")
#plt.xlim([1, 100])
#plt.ylim([-0.1, 1.1])
plt.savefig('box_window signal.png')
```

```
#FFT
#A = fft(windowed_sig , 2048) / 25.5
A = fft(windowed_sig , 1024) / 50
```

```
mag = np.abs(fftshift(A))
```

```
freq = np.linspace(-0.5, 0.5, len(A))
```

```
with np.errstate(divide='ignore', invalid='ignore'):
```

```
    response = 20 * np.log10(mag)
```

```
    response = np.clip(response, -100, 100)
```

```
plt.figure(9)
```

```
plt.plot(freq, response)
```

```
plt.title("Frequency response of the Box car window signal")
```

```
plt.ylabel("Magnitude [dB]")
```

```
plt.xlabel("Normalized frequency [cycles per sample]")
```

```
plt.axis('tight')
```

```
plt.savefig('Frequency response of the Box car window signal.png')
```


REFERENCES:

1. <https://www.sciencedirect.com/topics/engineering/periodic-signal>
2. <https://www.tutorialspoint.com/what-is-a-sinusoidal-wave-signal-definition-and-importance>
3. <https://vru.vibrationresearch.com/lesson/windowing-process-data/>
4. https://en.wikipedia.org/wiki/Fast_Fourier_transform
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6. <https://studip.hs-wismar.de/dispatch.php/course/overview?cid=b3bb2f3c71d55a6ec368d250f067a73f>
--"Dr.-Ing. Toralf Renkwitz"(studip code)